

Cambridge IGCSE™

BIOLOGY**0610/31**

Paper 3 Theory (Core)

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MARK SCHEME

Maximum Mark: 80

Published

This mark scheme contains the instructions and marking guidance that our examiners used to mark candidate responses for this component.

Before they start marking, examiners complete a standardisation process. They review a range of candidate responses and discuss the mark scheme in detail to agree which answers can and cannot be accepted. Examiners award marks where it is clear that candidate responses have met the requirements of the mark scheme. These requirements may vary between different components or different versions of the same component, depending on the exact requirements of the question and the candidate responses seen during the standardisation process.

Sometimes, our mark schemes may allow marks to be awarded to responses which contain elements that are factually incorrect or for content not covered by the syllabus. We do this in response to the demands of a specific question to support candidates and make sure that our assessments are fair. However, these allowances should not be used as a guide for teaching and learning.

We cannot discuss the mark scheme with schools, and we recommend that schools read the mark scheme together with the assessment material and the Principal Examiner Report for Teachers.

This document consists of **14** printed pages.

General marking principles

All examiners must apply these marking principles when marking candidate responses.

1	Examiners must award marks for each response in line with: - the specific content defined in the mark scheme - the specific skills defined in the mark scheme or in the level descriptions - the standard of response required as exemplified by the standardisation scripts.
2	Examiners must award whole marks (not half marks, or other fractions).
3	Examiners must award marks positively . This means that: - marks are not deducted for errors or omissions - marks are awarded for correct/valid answers as defined in the mark scheme and also for other valid answers which go beyond the scope of the syllabus and mark scheme referring to your team leader as appropriate - the quality of spelling, punctuation and grammar are judged only when the mark scheme indicates that these features are specifically assessed in the question. However, the meaning of all responses must be unambiguous.
4	Examiners must consistently apply the requirements of the mark scheme and any rules for what to do when candidates have not followed instructions correctly.
5	Examiners must use the full range of marks defined in the mark scheme, unless limited by the quality of the candidate responses seen.
6	Examiners must award marks according to the mark scheme and must not award marks with grade thresholds or grade descriptions in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Annotations guidance for centres

Examiners use a system of annotations as a shorthand for communicating their marking decisions to one another. Examiners are trained during the standardisation process on how and when to use annotations. The purpose of annotations is to inform the standardisation and monitoring processes and guide the supervising examiners when they are checking the work of examiners within their team. The meaning of annotations and how they are used is specific to each component and is understood by all examiners who mark the component.

We publish annotations in our mark schemes to help centres understand the annotations they may see on copies of scripts. Note that there may not be a direct correlation between the number of annotations on a script and the mark awarded. Similarly, the use of an annotation may not be an indication of the quality of the response.

The annotations listed below were available to examiners marking this component in this series.

Annotations

Annotation	Meaning
	correct point or mark awarded
	incorrect point or mark not awarded
	information missing or insufficient for credit
	allow or accept
	incorrect or insufficient point ignored while marking the rest of the response
	contradiction in response, mark not awarded
	benefit of the doubt given
	error carried forward applied
	point has been noted, but no credit has been given or blank page seen
	correct awarding one mark from marking point or marking group 1. similar numbered ticks are used for marking point or marking groups 2, 3, 4 etc.

Annotation	Meaning
	pages are linked together
	used to highlight part of the response
	used to highlight parts of an extended response
	used to highlight parts of an extended response
	Point already given
	Maximum mark reached
	Key point attempted / working towards marking point / incomplete answer / response seen but not credited / blank page seen
	Maximum number of marks for a marking point has been awarded.

Mark Scheme Abbreviations:	
;	separates marking points
/	alternative responses for the same marking point
R	reject the response
A	accept the response
I	ignore the response
ecf	error carried forward
AVP	any valid point
ora	or reverse argument
AW	alternative wording
<u>underline</u>	actual word given must be used by candidate (grammatical variants excepted)
()	the word / phrase in brackets is not required but sets the context
max	indicates the maximum number of marks that can be given
MP	marking point

Question	Answer	Marks	Guidance												
1(a)(i)	<i>Bombina</i> ;	1													
1(a)(ii)	animal ;	1													
1(b)	<table><tr><th>feature</th><th>amphibian</th><th>reptile</th></tr><tr><td>scales</td><td></td><td>✓</td></tr><tr><td>lungs</td><td>✓</td><td>✓</td></tr><tr><td>laying eggs</td><td>✓</td><td>✓</td></tr></table> ;;;	feature	amphibian	reptile	scales		✓	lungs	✓	✓	laying eggs	✓	✓	3	one mark for each correct row
feature	amphibian	reptile													
scales		✓													
lungs	✓	✓													
laying eggs	✓	✓													
1(c)	(presence of a) backbone / AW ;	1													
1(d)	<i>any two from:</i> birds ; mammals; fish ;	2													

Question	Answer	Marks	Guidance
2(a)(i)	2.28 ; 0.32 ; increases ;	3	ecf from incorrect MP1 used in MP2
2(a)(ii)	<i>any one from:</i> temperature affects (the rate of), water uptake / transpiration / evaporation ; increased temperature increases, transpiration / water uptake / evaporation / ora ;	1	
2(b)(i)	A – transport of, water / mineral ions OR support ; B – photosynthesis / absorbs light ;	2	

Question	Answer	Marks	Guidance
2(b)(ii)	X – evaporation ; Y – diffusion ;	2	
2(c)(i)	<i>any two from:</i> (root) hair (cell) / epidermal ; (root) cortex (cell) ; AVP ;	2	either order e.g. xylem
2(c)(ii)	solvent ;	1	

Question	Answer	Marks	Guidance
3(a)(i)	D ; E ; germination, requires oxygen and water / does not require light ;	3	D and E in either order
3(a)(ii)	(suitable) temperature ;	1	
3(b)(i)	anther circled ;	1	
3(b)(ii)	stigma labelled with an X ;	1	
3(b)(iii)	<i>any three from:</i> (has) petals ; anther / stamen, positioned inside, flower / petals ; stigma / carpel, positioned inside, flower / petals ; contains a nectary ; larger / heavier / less, pollen ; sticky pollen AW ; AVP ;	3	A produces nectar e.g. scent / coloured petals or flower
3(c)	filament ; anther ;	2	either order

Question	Answer	Marks	Guidance
3(d)	ovule ;	1	A ovary / embryo sac

Question	Answer	Marks	Guidance				
4(a)	heterozygous ; homozygous; phenotype ;	3					
4(b)	no no yes ;;	2	3 correct = 2 marks 2 correct = 1 mark				
4(c)	<table><tr><td>AA</td><td>AA</td></tr><tr><td>Aa</td><td>Aa</td></tr></table> 50(%) ; ;	AA	AA	Aa	Aa	2	MP1 four correct offspring MP2 correct percentage ecf from incorrect MP1
AA	AA						
Aa	Aa						
4(d)(i)	nucleus ;	1					
4(d)(ii)	X and Y ;	1	either order				

Question	Answer	Marks	Guidance
5(a)	<i>any four from:</i> (in, beaker B / higher temperature) 1 dye particles are more spread out / AW ; 2 ref. to diffusion ; 3 particles / dye, move faster ; 4 ref. to random movement (of particles) ; 5 (particles) have, more / higher, kinetic energy ; 6 increased temperature increases the <u>rate</u> (of diffusion) ; 7 dye / particles, moves from high(er) concentration to low(er) concentration ;	4	No reference to particles or molecules = max 3 A ora for MP 1,3 and 5 A molecules for particles A KE for kinetic energy A move down a concentration gradient.
5(b)(i)	active (transport) ;	1	
5(b)(ii)	respiration ;	1	I aerobic / anaerobic
5(b)(iii)	mitochondria ;	1	ecf from 5(b)(ii) i.e. accept cytoplasm if anaerobic given in 5(b)(ii)

Question	Answer	Marks	Guidance
6(a)(i)	increase in pupil diameter / decrease in iris size / AW ;	1	A increase in pupil size / pupil dilation
6(a)(ii)	(decrease in) light (intensity) / moving from light to dark / AW ;	1	A dark
6(a)(iii)	<i>any two from:</i> increase in heart rate ; increase in breathing rate ; AVP ;	2	e.g. increase in blood glucose concentration / airways widen / blood vessels constrict / pain tolerance increases / increase blood pressure / increases in respiration

Question	Answer	Marks	Guidance
6(b)(i)	B ; A ; C ;	3	
6(b)(ii)	endocrine ;	1	A ductless
6(c)	Hormones are chemical substances. are responsible for reflex actions. are transported through neurones. have effects that last longer than nervous control. have specific target organs. ...	3	one mark for each correct line R each additional line

Question	Answer	Marks	Guidance
7(a)	decreases blood <u>glucose</u> (concentration) ;	1	
7(b)(i)	genetic modification ;	1	
7(b)(ii)	<i>any two from:</i> rapid reproduction rate ; can produce complex molecules ; AVP ;	2	e.g. lack of ethical concerns / presence of plasmids / cheaper production (than animals)
7(b)(iii)	<i>any two from:</i> cell wall ; circular DNA ; plasmids ; AVP ;	2	A loop of DNA
7(c)(i)	<i>any three from:</i> food supply / nutrients ; competition ; disease ; toxins / waste ; AVP ;	3	e.g. oxygen supply / temperature / pH
7(c)(ii)	A – lag ; C – stationary ;	2	

Question	Answer	Marks	Guidance
8(a)	<i>any four from:</i> hunted by humans / poaching ; increased predation ; lack of food / water ; (named) destruction of their habitat AW ; pollution (of their habitat) ; competition (from other organisms) ; introduced / foreign, species ; climate change / named example of climate change ; disease ; AVP ;	4	A deforestation / (named) natural disasters e.g. global warming / sea levels rise e.g. infertility / lack of mates
8(b)(i)	captive breeding ;	1	I selective breeding A IVF / AI
8(b)(ii)	1150 (%) ;;	2	MP1 correct selection of data and calculation of change in number of individuals i.e. $250 - 20 = 230$ MP2 correct calculation of the percentage increase i.e. $(230 \div 20) \times 100 = 1150$ 1150 with no working = 2 marks
8(c)	They can produce fertile offspring. ticked ; They respire. ticked ;	2	box 1 box 4 R each additional tick
8(d)	(a unit containing the) community (of organisms) and their environment ; interacting (together) ;	2	A MP1 <u>all</u> the living things and their environment / <u>all</u> the living and non-living things